Predicting Coronary Heart Disease from Lifestyle Factors: A Robust Pipeline with Class-Imbalance Handling

Lila Reinhardt, Isaure ..., Partner ...
Department of Computer Science / Life Sciences Engineering, EPFL, Switzerland

***Abstract*—Cardiovascular diseases (CVD), notably myocardial infarction and coronary heart disease (MICHD), represent a leading cause of mortality worldwide. In this work, we develop an end-to-end machine learning classification system using pure NumPy to predict MICHD occurrence from the Behavioral Risk Factor Surveillance System (BRFSS) dataset (over 300,000 respondents, 321 raw features). Our approach focuses on domain-informed preprocessing—specifically detecting survey sentinel missingness codes without manual codebook inspection—coupled with stratified handling of severe class imbalance (∼9:1 ratio). We rigorously benchmark the six standard optimization methods and demonstrate that regularized logistic regression, augmented with class undersampling and threshold tuning on out-of-fold cross-validation scores, achieves an F_1-score of 0.395 (Accuracy 81.2%), significantly outperforming naive baselines ($F_1 = 0.0$).**

I. Introduction

Cardiovascular diseases (CVD), particularly myocardial infarction and coronary heart disease (MICHD), are primary targets for preventive healthcare. The Behavioral Risk Factor Surveillance System (BRFSS) collected by the CDC records annual health behaviors, chronic conditions, and clinical indicators. Machine learning models trained on this survey can identify at-risk profiles early, enabling timely intervention.

However, clinical survey datasets present three major hurdles for linear modeling:

1) **High-dimensional sparsity and sentinel encoding:** The dataset comprises $N = 328{,}135$ observations across $D = 321$ initial features. Many survey questions employ arbitrary numeric codes (e.g., 7, 9, 77, 999) denoting "refused" or "don't know", which artificially distort distance metrics if treated as continuous values.
2) **Acute class imbalance:** Only ∼8.8% of survey respondents reported an MICHD event. Standard empirical risk minimization easily converges to the trivial majority classifier, attaining 91.2% accuracy while failing to detect a single affected individual.
3) **Strict operational constraints:** All processing, linear solvers, optimization routines, and cross-validation pipelines must be implemented strictly using primitive NumPy operations without external learning libraries.

This report details our modular pipeline, an automated gap-detection heuristic for sentinel codes, an empirical evaluation of six baseline methods, and an ablation study assessing data-cleaning and balancing strategies.

II. Models and Methods

A. Underlying Mathematical Models

We consider binary classification where inputs are standardized design matrices $\tilde{X} \in \mathbb{R}^{N\times(D+1)}$ (including an intercept column of ones) and binary targets $y \in \{-1, +1\}$, mapped to $y \in \{0, 1\}$ for probabilistic models.

Linear Ridge Model: Assumes a continuous output $\hat{y} = \tilde{X}w$ minimizing regularized Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(w) = \frac{1}{2N}\|y - \tilde{X}w\|_2^2 + \lambda\|w\|_2^2. \tag{1}$$

Logistic Regression Model: Models the posterior probability $p(y_i = 1 \mid \tilde{x}_i) = \sigma(\tilde{x}_i w) = (1 + e^{-\tilde{x}_i w})^{-1}$. The optimal weights minimize the penalized negative log-likelihood (NLL):

$$\mathcal{L}_{\text{NLL}}(w) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \ln\sigma(\tilde{x}_i w) + (1 - y_i)\ln(1 - \sigma(\tilde{x}_i w))] + \lambda\|w\|_2^2. \tag{2}$$

B. Optimization Methods

In accordance with course specifications, we implemented six standalone algorithms in `implementations.py`:

- **Least Squares:** Closed-form solution via Cholesky/normal equations: $w = (\tilde{X}^T\tilde{X})^{-1}\tilde{X}^T y$, implemented using `np.linalg.solve`.
- **Ridge Regression:** Closed-form L_2-penalized regression: $w = (\tilde{X}^T\tilde{X} + 2N\lambda I)^{-1}\tilde{X}^T y$.
- **MSE Gradient Descent (GD) and SGD:** Iterative first-order updates $w^{(t+1)} = w^{(t)} - \gamma\nabla\mathcal{L}(w^{(t)})$, with mini-batch size $B = 1$ for SGD.
- **Logistic Regression (GD):** Iterative updates using exact gradient $\nabla\mathcal{L} = \frac{1}{N}\tilde{X}^T(\sigma(\tilde{X}w) - y)$.
- **Regularized Logistic Regression (GD):** Incorporates weight penalty gradient $2\lambda w$ (excluding intercept).

C. Automated Survey Sentinel Code Cleaning

Rather than manually inspecting 300 pages of the CDC codebook, we implemented an automated statistical heuristic (`clean_special_codes_auto`). For each feature $j \in \{1, \ldots, D\}$ computed solely on the training fold:

- **Gap Analysis:** Candidate sentinel values $c \geq 77$ are classified as invalid non-responses if $c > 1.5 \times \max(X_{\cdot,j} \setminus \{c\})$. This reliably catches codes like 77 (refused) for bounded attributes (e.g., drinks per month ≤ 30).

- **Categorical Refusal:** Values in $\{7, 9\}$ are marked invalid if the remaining unique entries satisfy $\max(X_{\cdot,j} \setminus \{c\}) \leq 5$, characteristic of ordinal scales.

Identified values are replaced by `NaN`. Columns exceeding a missingness threshold ($> 50\%$ NaNs) or near-zero variance ($\sigma^2 < 10^{-8}$) are discarded. Missing values are imputed using column-wise training medians, followed by zero-mean unit-variance standardization and bias term appending.

D. Class-Imbalance Strategy and Threshold Tuning

Due to the $\sim$9:1 class disparity, we employ training-fold majority undersampling (`with_undersampling`), enforcing a balanced 1:1 negative-to-positive ratio per training fold, leaving validation folds untouched. Furthermore, since maximum likelihood produces uncalibrated marginal probabilities under altered class priors, we tune the decision threshold $\tau \in [0, 1]$ directly on out-of-fold cross-validation scores using a fast $O(N \log N)$ cumulative sum scan to maximize the F_1-score:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3}$$

III. Experimental Results and Discussion

All experiments are conducted using 5-fold stratified cross-validation (`seed=1`). Performance is evaluated via Accuracy, Precision, Recall, and F_1-score.

TABLE I
BASELINE COMPARISON OF OFFICIAL METHODS (5-FOLD CV).

Method	Accuracy	Precision	Recall	F_1-Score
Majority Baseline ($\hat{y} = -1$)	0.912	0.000	0.000	0.000
Least Squares	0.884	0.281	0.219	0.246
Ridge ($\lambda = 10^{-2}$)	0.885	0.285	0.214	0.244
MSE GD ($\gamma = 0.01, 100$ it.)	0.880	0.264	0.228	0.244
Logistic Reg ($\gamma = 0.05, 300$ it.)	0.893	0.320	0.201	0.247
Reg. Logistic ($\lambda = 10^{-4}$)	0.894	0.325	0.205	0.251
Reg. Logistic + Undersampling	**0.812**	**0.298**	**0.584**	**0.395**

A. Baseline Comparison

Table I contrasts the six required methods against our optimized pipeline. Without imbalance intervention, all models attain high accuracy ($\sim$89–91%) but dismal F_1-scores (≤ 0.251), suffering from severe false-negative rates ($Recall \approx 20\%$). Closed-form Ridge and Least Squares converge instantly, while Gradient Descent requires carefully tuned learning rates ($\gamma \approx 0.01$) to prevent divergence given non-convexities introduced by unscaled survey artifacts.

B. Ablation Study

To isolate the marginal utility of each component, we performed an ablation study summarized in Table II.

The ablation confirms that replacing sentinel codes with median imputation is crucial: it prevents codes like 999 from acting as spurious extreme regressors, raising F_1 from 0.182 to 0.251. Combining balanced training with decision-threshold calibration yields the most substantial leap in F_1 (+0.144), balancing precision and clinical sensitivity.

TABLE II
ABLATION STUDY OF PIPELINE COMPONENTS (REG. LOGISTIC).

Configuration	Accuracy	Recall	F_1-Score
Raw Data (Mean Impute, No Cleaning)	0.908	0.124	0.182
+ Automated Sentinel Cleaning	0.894	0.205	0.251
+ Undersampling (1:1 Ratio)	0.771	0.762	0.371
+ Optimal Threshold Tuning ($\tau^* = 0.58$)	**0.812**	**0.584**	**0.395**

IV. Conclusion and Reproducibility

By pairing careful data-sanitization heuristics with balanced gradient updates, we demonstrated that simple regularized linear models can achieve competitive predictive performance on massive, noisy survey datasets without non-linear libraries. Our complete experimental workflow, including baseline reproduction and the final competition submission generator, is fully self-contained in our accompanying `run.py` repository.

REFERENCES

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